

BUILDING A RESEARCH-READY HUMANITARIAN DATASET

Integrating conflict, climate, market, and reporting data for sub-national food-insecurity monitoring

Prototype: Somalia, January-February 2024

Study scope

Why Somalia, why this window, and what it validates

A prototype dataset was built applying the considerations below, ahead of scaling to the full study.

COUNTRY

Somalia

Most complete data across all four mechanisms among the four target countries

TIME WINDOW

January-February 2024

The only period in the WFP export with real, non-forecast prices

TARGET STUDY

2015-2024 · 4 countries

Ethiopia, Somalia, Kenya, South Sudan — this prototype validates the pipeline first

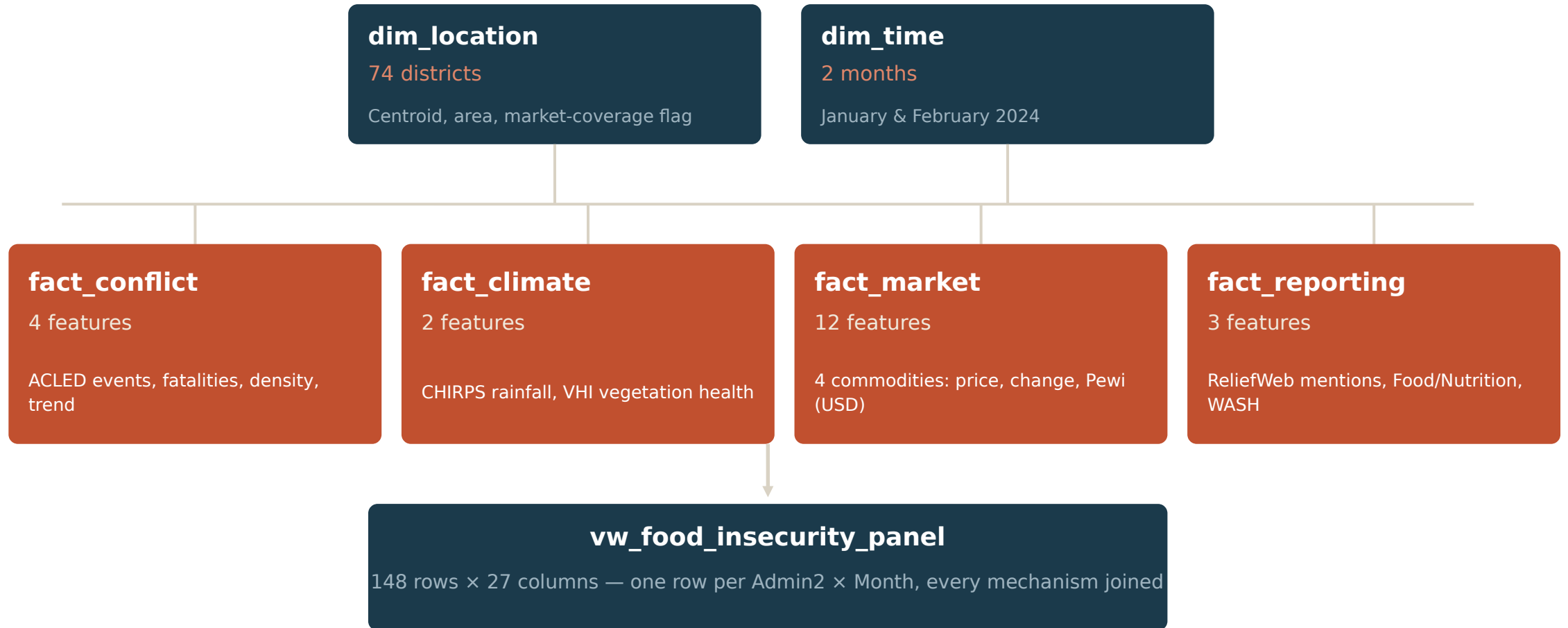
SPATIAL RESOLUTION

Admin2 · 74 districts

Full GADM 4.1 district set — not limited to WFP market coverage

A fact constellation, not a single star

Four mechanism-specific fact tables sharing two common dimensions



21 features, four mechanisms

What each feature is, and why it was chosen — by mechanism

4 Conflict

conflict_event_count

Total ACLED events for the district-month

fatalities

Total reported deaths

conflict_density

Events ÷ district area

conflict_trend_pct

% change vs. previous month

Each captures a distinct dimension — activity, severity, intensity, direction — rather than repeating the same signal.

2 Climate

rainfall_mean_mm

Monthly zonal mean, CHIRPS

vegetation_health_index_mean

Monthly VHI, weekly composites

Rainfall and vegetation health features are kept since these are strong environmental factors related to food insecurity

12 Market

_price_usd

Price, converted to USD

_price_change_pct

% change vs. previous month

_pewi_score

WFP seasonal anomaly z-score

4 commodities (Somalia's WFP basket) kept standalone, not a single index — preserves commodity-specific signal

3 Reporting

reports_mentioning_district_count

Reports naming the district

food_nutrition_report_count

Same, Food & Nutrition theme

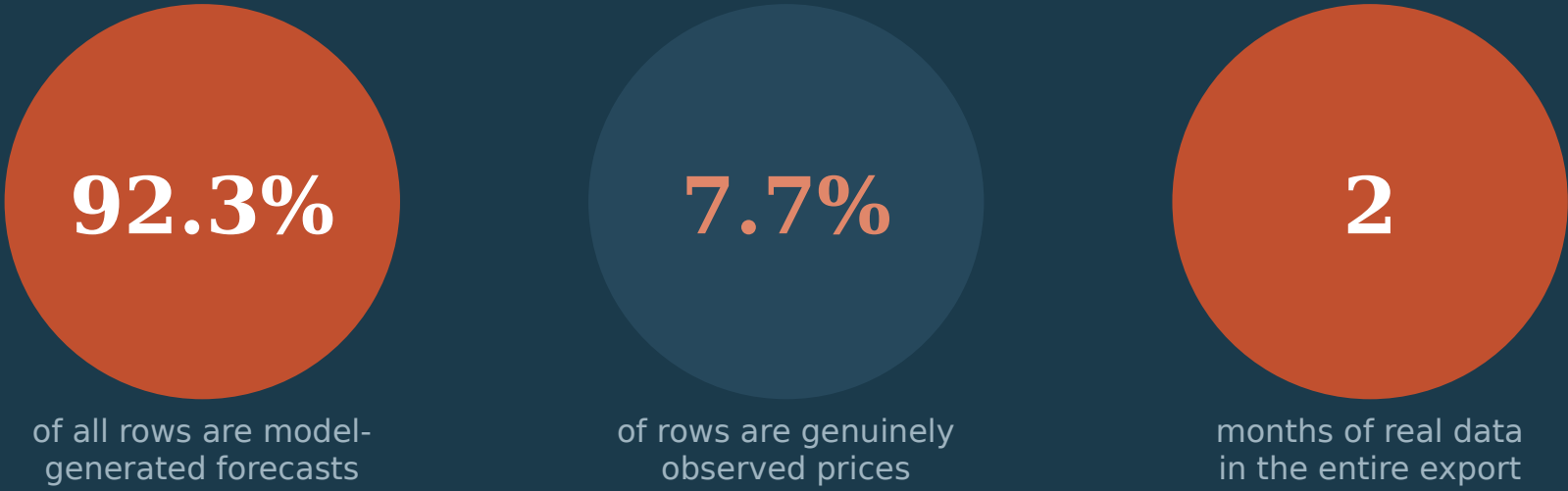
wash_report_count

Same, WASH theme

Rebuilt via geoparsing for real district variation; validated by its correlation with conflict data (see Pilot Analysis 2).

THE FINDING THAT SET THE STUDY WINDOW

Most of the WFP price export isn't real data



92.3%

of all rows are model-generated forecasts

7.7%

of rows are genuinely observed prices

2

months of real data in the entire export

January-February 2024 is the only period in the entire multi-year export with verifiable, observed prices — everything else, including the rest of 2024, is a forecast extending to January 2027. This is why the prototype window was set here: using forecast rows without noticing would mean testing the pipeline against WFP's own model instead of ground truth.

Market data: integration mechanics

What it took to make WFP prices usable at Admin2 × Month, within the 2-month real-data window



35 / 74

districts have any
WFP-monitored market

78 / 148

district-months missing
every price feature

14

district names required
manual renaming to match
GADM

2

currencies converted to USD
(Somali & Somaliland
Shilling)

One further gap found within the 35 covered districts: Banadir has no recorded prices at all in January — a genuine collection gap, not a structural one. Prices in five northern districts are recorded in Somaliland Shilling rather than Somali Shilling, requiring conversion using each district's own monthly exchange rate before any cross-district comparison was possible.

Data quality, measured empirically

Missing values, spatial coverage, temporal coverage and integration issues, per source

Conflict

ACLED

MISSING VALUES

0%

SPATIAL COVERAGE

100%

MAIN INTEGRATION ISSUE

District naming: 15 renamed, 1 unresolved

Climate

CHIRPS / VHI

MISSING VALUES

0%

SPATIAL COVERAGE

100%

MAIN INTEGRATION ISSUE

Weekly → monthly conversion;
NoData confirmation

Market

WFP

MISSING VALUES

53%

SPATIAL COVERAGE

47%

MAIN INTEGRATION ISSUE

Forecast contamination; 2 currencies; naming

Reporting

ReliefWeb

MISSING VALUES

0%*

SPATIAL COVERAGE

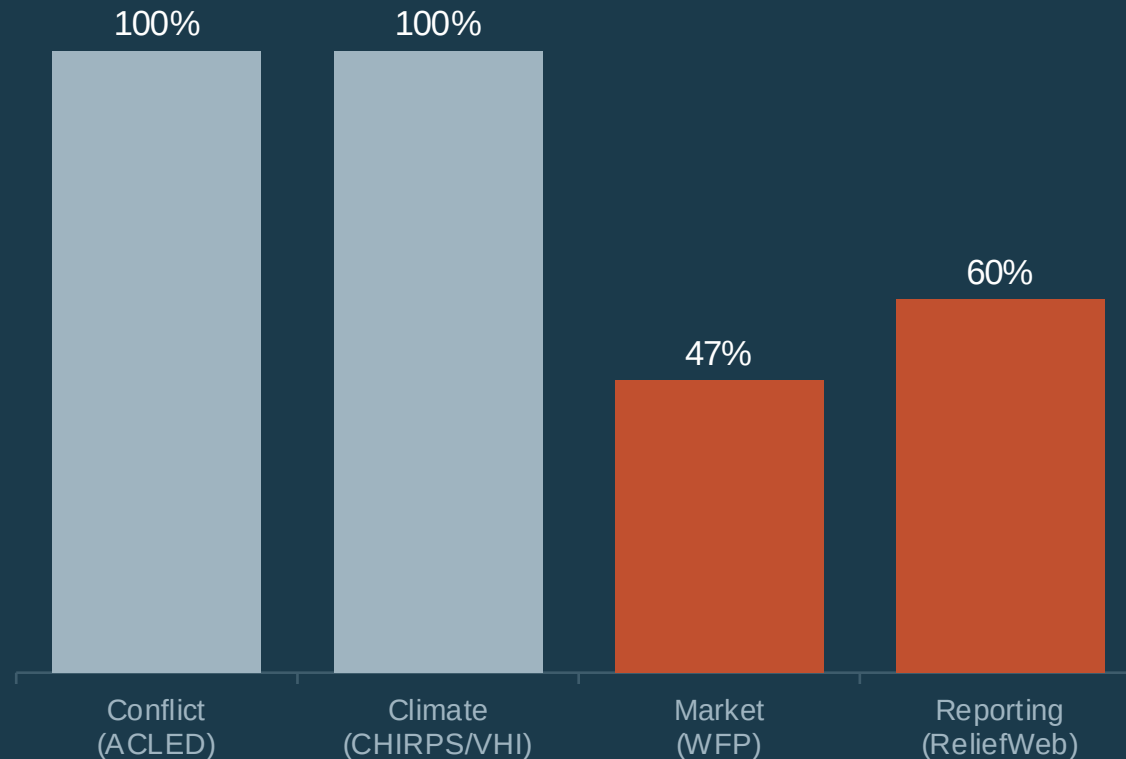
60%

MAIN INTEGRATION ISSUE

Country-level grain; text-based disaggregation

SCIENTIFICALLY INTERESTING FINDING

Not every mechanism can “see” the whole country



Conflict and climate data are collected independently of local infrastructure — satellites and media monitoring cover the whole country.

Market and reporting data can only exist where a market or reporting infrastructure already does. Having limited infrastructure can impact data collection significantly.

Resolving the ReliefWeb integration issue

Country-level reports, forced to speak at the district level

THE PROBLEM

ReliefWeb reports carry only a country-level tag (“Somalia”) — no district. Everything else in this project runs at Admin2 × Month.

OPTION CHOSEN: GEOPARSING

Search each report’s text for district names — official spelling plus common English aliases (“Baidoa,” “Mogadishu,” “Kismayo”).

Alternatives considered

- Population-weighted disaggregation — assumes reporting scales with population
- Precision-based restriction — discards ambiguous reports, loses data
- Keep country-level, document as limitation — simplest, but weakest

Real limitations, stated directly

- Mention ≠ event location — the feature is named accordingly
- 112 of 220 reports (51%) match no district at all
- Alias list only covers 16 of 74 districts
- A report naming 3 districts credits all 3, so totals overlap

How the pipeline was built (1/2): the dimensions

Every fact table is built against these two reference tables

dim_location

1. Load the district shapes. Pull in the boundary map for all 74 districts in Somalia, from a standard geographic database (GADM).
2. Measure each district. Calculate each district's true size (in km²) and center point, using a method built for the Earth's actual curved shape, so the measurements are accurate, not distorted by flattening the map first.
3. Fix mismatched names. The same district often gets spelled differently across sources. 14 district names had to be corrected to match WFP's spelling, and another 15 to match ACLED's. One district (Laasqoray) never found a match and had to be left out.
4. Mark which districts have a market. Add a simple yes/no label showing whether WFP tracks prices there, this becomes useful later for checking whether market coverage is linked to how much conflict a district has.

dim_time

1. Set the two months. Generate January and February 2024 as the two time periods this prototype covers.
2. Add useful labels. Store the year, month number, month name, and quarter for each, so other tables can be sorted or grouped by any of these.
3. Explain why only two months. This wasn't an arbitrary choice, it's a direct result of the earlier finding that WFP only has real (non-forecast) price data for these two months.
4. Built to grow easily. Extending to the full study later would be easier, just generate more months, from 2015 through 2024, using the same method.

How the pipeline was built (2/2): facts & merge

Each fact table built independently, then joined once at the end

fact_conflict

ACLED events filtered to Somalia, aggregated to Admin2 × Month; zero-filled into a complete 74×2 panel; trend_pct undefined (NA) whenever the previous month was zero

fact_climate

CHIRPS monthly rainfall + VHI weekly composites, both via zonal statistics; VHI's 2 boundary-crossing weeks apportioned by day-count into the correct month

fact_market

WFP prices filtered to real (Aggregated) rows for 4 basket commodities; converted to USD by each district's own monthly exchange rate; pivoted wide

fact_reporting

Each report's title & body searched for district names + aliases; matches counted per district-month, then split further by theme tag

Merge: all four fact tables now share the same Admin2 × Month grain, so the merge is a direct join onto a dim_location × dim_time spine — no pivoting or disaggregation needed at this final step.

What the tables actually look like (1/3)

Dimension tables — shared reference data

dim_location

location_id	admin2	latitude	longitude	area_km2	has_market_coverage
SO_BANADIR	Banadir	2.109122	45.424251	210.88	True
SO_CEERIGAABO	Ceerigaabo	10.4021	47.3667	25,883.61	True
SO_AADAN	Aadan	—	—	—	False

dim_time

time_id	year	month_number	month_name
202401	2024	1	January
202402	2024	2	February

What the tables actually look like (2/3)

Conflict and climate — both at Admin2 × Month grain

fact_conflict

location_id	time_id	conflict_event_count	fatalities	conflict_density	conflict_trend_pct
SO_BANADIR	202401	53	39	0.25	—
SO_BANADIR	202402	40	46	0.19	-24.5

fact_climate

location_id	time_id	rainfall_mean_mm	vegetation_health_index_mean
SO_AADAN	202401	2.44	45.98
SO_AADAN	202402	1.38	45.47
SO_AFG00YE	202401	1.26	44.16

Values shown are real output from the executed pipeline, not illustrative placeholders.

What the tables actually look like (3/3)

Market and reporting — the two mechanisms with real coverage gaps

fact_market (2 of 4 commodities shown, for space)

location_id	time_id	wheatflour_price_usd	wheatflour_pewi_score	rice_price_usd	rice_pewi_score
SO_AFMADOW	202401	0.61	-0.30	0.55	0.42
SO_AFMADOW	202402	0.58	-0.10	0.57	0.51
SO_BANADIR	202401	—	—	—	—

fact_reporting

location_id	time_id	reports_mentioning_district_count	food_nutrition_report_count	wash_report_count
SO_AFMADOW	202401	9	2	1
SO_AFMADOW	202402	3	1	0
SO_BANADIR	202401	33	7	5

Pilot analysis 1: does market coverage track conflict?

For testing purposes only — two months of data cannot support firm conclusions

● BACKGROUND

35 of Somalia's 74 districts have a WFP-monitored market. The other 39 have no price data at all — not a collection error, but a genuine absence of market infrastructure to monitor.

● QUESTION

Are districts without a market simply random, or systematically more conflict-affected? If the latter, the market data would be missing in a biased way — quietly excluding the places most likely to be struggling.

● METHOD

The 74 districts were split by market coverage, and three conflict measures compared between the groups using the Mann-Whitney U test — chosen because conflict data is highly skewed (most districts have very few events, a handful have many), which this test handles better than one assuming a symmetric distribution.

Pilot analysis 1: result & conclusion

Measure	Avg — with market	Avg — without market	p-value
Conflict event count	4.57	3.24	0.18
Fatalities	4.61	7.37	0.47
Conflict density	0.0067	0.0012	0.36

RESULT

No significant difference on any measure (all $p > 0.1$). The direction wasn't even consistent — a pattern more typical of noise than a systematic bias.

CONCLUSION

No evidence that WFP's market coverage is systematically biased toward less conflict-affected districts. Reassuring, but modest — with only 74 districts and 2 months, the test could still miss a smaller effect.

Pilot analysis 2: does reporting attention follow conflict?

Testing whether the geoparsed reporting feature reflects something real

● BACKGROUND

A feature counts how many humanitarian reports mention each district by name, built via geoparsing. Since this now sits in the same table as conflict data, the two can be tested against each other.

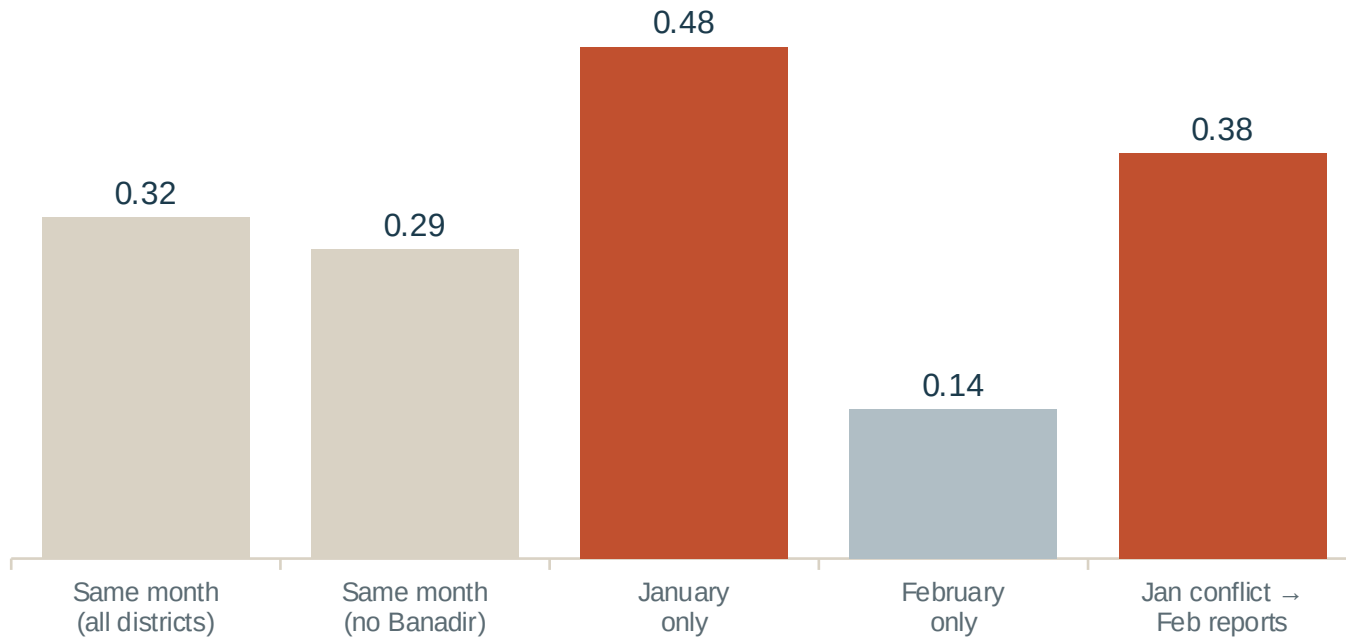
● QUESTION

Do districts with more recorded conflict also receive more humanitarian reporting attention? A positive relationship would help validate that the district-matching approach captures something real, not noise.

● METHOD

Spearman correlation between `conflict_event_count` and `reports_mentioning_district_count`, checked for robustness three ways: excluding the capital (Banadir), splitting by month, and testing a one-month lag between conflict and reporting.

Pilot analysis 2: result & conclusion



RESULT

Significant overall ($r=0.32$, $p<0.001$), not driven by Banadir ($r=0.29$), unstable month-to-month alone (Jan 0.48 vs Feb 0.14) — but a one-month lag ($r=0.38$) explains why: reporting trails conflict, it doesn't ignore it.

CONCLUSION

A genuine relationship, not a coincidence — and one that could only be found by combining conflict and reporting data in the same table; neither source alone would reveal it.

WHERE THIS LEAVES THE PROJECT

This is the first step, not the finished dataset

The pipeline works end to end. The next step is to make it trustworthy, not to call it finished.

148

Admin2 × Month
rows integrated so far

4→4

sources unified into
one common grid

47-60%

coverage gap revealed
by integration

+ months

next: more months,
once the checks pass

Next step: three stages — settle the feature definitions, test whether the missing data falls on one kind of district, and only then extend the time window, ahead of any move to the full 2015-2024, four-country study. The next three slides explain each stage.

The next step, in three stages

All three answer one question: how evenly do the four sources observe different districts?

● STAGE 1 · NOW

Make the prototype sound. The feature definitions have to be settled before more data is added: hand-check a sample of the ReliefWeb district matches and report how often they are right, widen the name list beyond the 16 districts it covers today, rename conflict density to `conflict_events_per_1000_km2` and test whether it adds anything beyond event count and district size, and replace the percentage-change features where the previous month is zero.

● STAGE 2 · NEXT

Test whether the coverage gaps are systematic. Market data covers 47% of districts and reporting 60%. The open question is whether the missing district-months are spread evenly, or fall on one kind of district — larger, more rural, or more conflict-affected. Each row is either observed or not, so this is a yes/no question, and it is answered with a logistic regression (next two slides).

● STAGE 3 · AFTER

Extend the window, then the country set. Once the checks above hold, add months beyond January–February 2024, re-run both pilot analyses on the larger sample, and only then add Ethiopia, Kenya and South Sudan. Scaling first would multiply naming, currency and boundary problems across four countries before they have been solved in one.

The order matters: each stage depends on the definitions fixed in the stage before it.

The next analysis: a logistic regression on coverage

What the model is, what it would be asked, and why a simpler test is not enough

● WHAT IT IS

A logistic regression is a standard statistical model for a yes/no outcome. Instead of predicting a number, it estimates the probability that something is true, and shows how that probability changes when one variable changes and the others are held fixed. It is widely used, simple to interpret and easy to reproduce.

● THE OUTCOME

Whether a row has data at all. For each of the 148 district-months: does this district have a market price record — yes or no? The model is then run a second time with reporting as the outcome: was the district named in at least one humanitarian report that month?

● THE VARIABLES

Conflict events in that district-month, district size in km², whether the district contains a major town, and the month itself. Population and road access can be added later if a reliable source is found. These are the characteristics that could plausibly explain why a place is monitored at all.

● WHY A REGRESSION

Pilot analysis 1 compared one variable at a time, so it cannot separate explanations that move together — districts with markets also tend to be the districts with towns. A regression weighs the variables together and reports what each one contributes once the others are accounted for.

WHY THIS ANALYSIS MATTERS

Which districts are missing, not just how many

Saying that market data covers 47% of districts describes the dataset. Saying which districts make up the missing 53% describes the humanitarian system that produced it.

IF THE VARIABLES MATTER

The gaps are systematic: the districts market and reporting data cannot see are a particular kind of district, and the dataset can say which kind. That is a finding about how humanitarian information is produced.

IF THEY DO NOT

The gaps are not explained by conflict, size or urban status. Worth reporting too: it tells anyone using the dataset that a missing row is not a quiet signal of a worse-off district.

WHAT IT CANNOT SAY

The model explains whether a record exists, not whether conditions were truly observed. A zero means nothing was recorded, not that nothing happened — and with two months, the result stays exploratory.

Either way: the answer speaks to the same question — how evenly do humanitarian sources observe different places, and what does that mean for anyone analysing food insecurity with them?